Diffraction

Ghatak §18.1–18.2, 18.6–18.8, 20.1–20.3, 20.6 (+ circular aperture) · pattern of the Nov 2024 / Nov 2025 papers, doubled

HOW THIS PAPER WORKS

A real paper has 12 short, 7 paragraph and 2 essay questions. This paper has **twice as many** (24 + 14 + 4), all from Module 3, covering every concept once as its main point.

Time	4 hours (or two sittings of 2 hours)
Maximum	120 marks

PYQ N25 Q5 = Nov 2025, question 5. "also" lists other years that asked the same thing. "adapted" = a 1-mark fill-in or True/False turned into a short answer.

Section A · Short answer

Answer all questions in two or three sentences. Each carries 2 marks. Ceiling 40.

1. How does a Fresnel diffraction differ from a Fraunhofer diffraction?

PYQ N25 Q5 **also N20 Q4**

2. State the Huygens–Fresnel principle.

3. Distinguish between interference and diffraction.

4. How does the single-slit Fraunhofer pattern change when (a) the slit is made narrower, (b) the wavelength is increased?

5. Where are the secondary maxima of a single slit pattern? Why are they not exactly midway between the minima?

6. Sound bends easily round buildings, but light does not. Explain.

7. "The diffraction pattern due to a circular aperture for a plane wave consists of concentric circular rings." True or false? Explain.

   PYQ N19 Q7 · adapted

8. To get more directionality for sound waves how will you select the diameter of the loud speaker?

   PYQ N19 Q12

9. What are missing orders in a double-slit diffraction pattern? State the condition.

10. "In N-slit Fraunhofer diffraction pattern, the principal maximum becomes sharper as N increases." True or false? Justify.

    PYQ N22 Q10 · adapted **also N19 Q6**

11. For N slits, how many minima and secondary maxima lie between two principal maxima? Why is the intensity of a principal maximum proportional to N^2?

12. Explain a diffraction grating.

PYQ N20 Q15

13. "Larger the grating element, larger will be the angular dispersion." True or false? Justify.

PYQ N19 Q9 · adapted

14. What is meant by overlapping spectra in the spectra of a grating?

PYQ N22 Q15

15. "The spectra obtained with a grating are comparatively pure than those with a prism." True or false? Explain.

PYQ N20 Q7 · adapted

16. Write down the equation for resolving power of a grating.

PYQ N24 Q8 also N22 Q5

17. State Rayleigh's criterion of resolution.

18. Find the highest order spectrum visible with a grating of 5000 lines per cm for light of wavelength 5000 Å.

19. For a plane wave front of monochromatic light, what is the difference in areas of nth and (n – 1)th Fresnel half period zones?

PYQ N25 Q6

20. Show that the resultant amplitude due to a whole wavefront is half that due to the first half-period zone.

21. Why is the centre of the shadow of a small circular disc bright?

22. Compare a zone plate with a convex lens.

23. What is the intensity at the edge of the geometrical shadow of a straight edge? Why?

24. What is the Cornu spiral? What do its two end points represent?

Section B · Paragraph / problem

Answer all questions in about half a page to one page. Each carries 5 marks. Ceiling 60.

25. Derive the equation for width of principal maxima of an N slit Fraunhofer diffraction.

PYQ N24 Q17

26. Obtain the intensity distribution in the Fraunhofer diffraction pattern of two slits. If the separation of the slits is three times their width, which orders are missing and how many fringes lie in the central envelope?

27. Derive the expression for the resolving power of a grating. A diffraction grating has 0.15 m of surface ruled with 6×10^5 lines per metre. What is its resolving power in the first order?

PYQ N22 Q29 also N25 Q15

28. A grating (with 15,000 lines per inch) is illuminated by sodium light. The grating spectrum is observed on the focal plane of a convex lens of focal length 10 cm. Calculate the separation between the D_1 and D_2 lines of sodium (5890 and 5896 Å).

PYQ N20 Q27

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29. Discuss on the resolving power of a microscope. Why oil immersion objectives are preferred in microscopes?

PYQ N19 Q19

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30. If the diameter of the pupil of the eye is 2 mm, find the minimum separation between two points to be resolved by eye. Given that the distance of the points from the eye is 30 m and the points are red in colour with wavelength 6×10^{-7} m.

PYQ N19 Q28

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31. Discuss the intensity distribution caused by Fraunhofer diffraction at a circular aperture.

PYQ N20 Q21 **also N22 Q22**

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32. Explain the construction of Fresnel's half period zones on a plane wave front. Find the area of each zone.

PYQ N19 Q20 **also N24 Q19**

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33. Light of wavelength 5000 Å from a distant source falls on a circular hole of radius 0.5 mm. Find the distances on the axis at which the intensity is maximum and minimum, nearest to the hole. Explain.

34. Explain the principle and construction of a zone plate.

PYQ N22 Q21

35. A zone plate has radii $r_n = 0.001\sqrt{n}$ m for λ = 4.5×10^{-7} m. Find the positions of foci.

PYQ N19 Q29 also N20 Q30

36. Show that a zone plate obeys a lens-like formula $\frac{1}{a} + \frac{1}{b} = \frac{1}{f}$ and obtain its focal length.

37. In the diffraction pattern of a straight edge, λ = 6000 Å and the screen is 120 cm from the edge. Find the distances of the first two maxima from the edge of the geometrical shadow (maxima at v = 1.22 and 2.34).

38. A slit 0.2 mm wide is illuminated normally by light of wavelength 5000 Å, and the pattern is seen in the focal plane of a lens of focal length 20 cm. Find the positions of the first two minima and the width of the central maximum.

Section C · Essay

Answer any two, in about two pages each. Each carries 10 marks.

39. Discuss Fraunhofer type of diffraction produced by a narrow slit illuminated by monochromatic light. Obtain the positions of the maxima and minima and draw a diagram to indicate the distribution intensity of light in the diffraction pattern.

PYQ N20 Q33 · also N19 Q33 · N22 Q33 · N24 Q21

40. Derive the intensity distribution for diffraction due to N identical slits.

PYQ N25 Q21

41. Explain the rectilinear propagation of light on the basis of Fresnel's half period zones.

PYQ N22 Q34

42. Discuss Fresnel diffraction at a straight edge. Introduce Fresnel's integrals and the Cornu spiral, obtain the intensity distribution, and describe the pattern in the shadow and in the illuminated region.

Coverage check

TOPIC	QUESTIONS
T1 Diffraction, Fresnel vs Fraunhofer	1, 2, 3, 6
T2 Single slit intensity	39
T3 Minima, secondary maxima, width	4, 5, 38
T4 Circular aperture, resolution	7, 8, 29, 30, 31
T5 Two slits, missing orders	9, 26
T6 N slits	10, 11, 40
T7 Width of principal maxima	25
T8 Grating, dispersion, overlap	12, 13, 14, 15, 18, 28
T9 Resolving power, Rayleigh	16, 17, 27
T10 Half-period zones	19, 20, 32
T11 Rectilinear propagation, hole, disc	21, 33, 41
T12 Zone plate	22, 34, 35, 36
T13 Straight edge, Cornu spiral	23, 24, 37, 42

All 32 Module 3 past questions (2019–2025) appear, directly or as "also".